

Village pond siting report

contours_1m.kml · analysed 2026-08-28 · [Full](#)

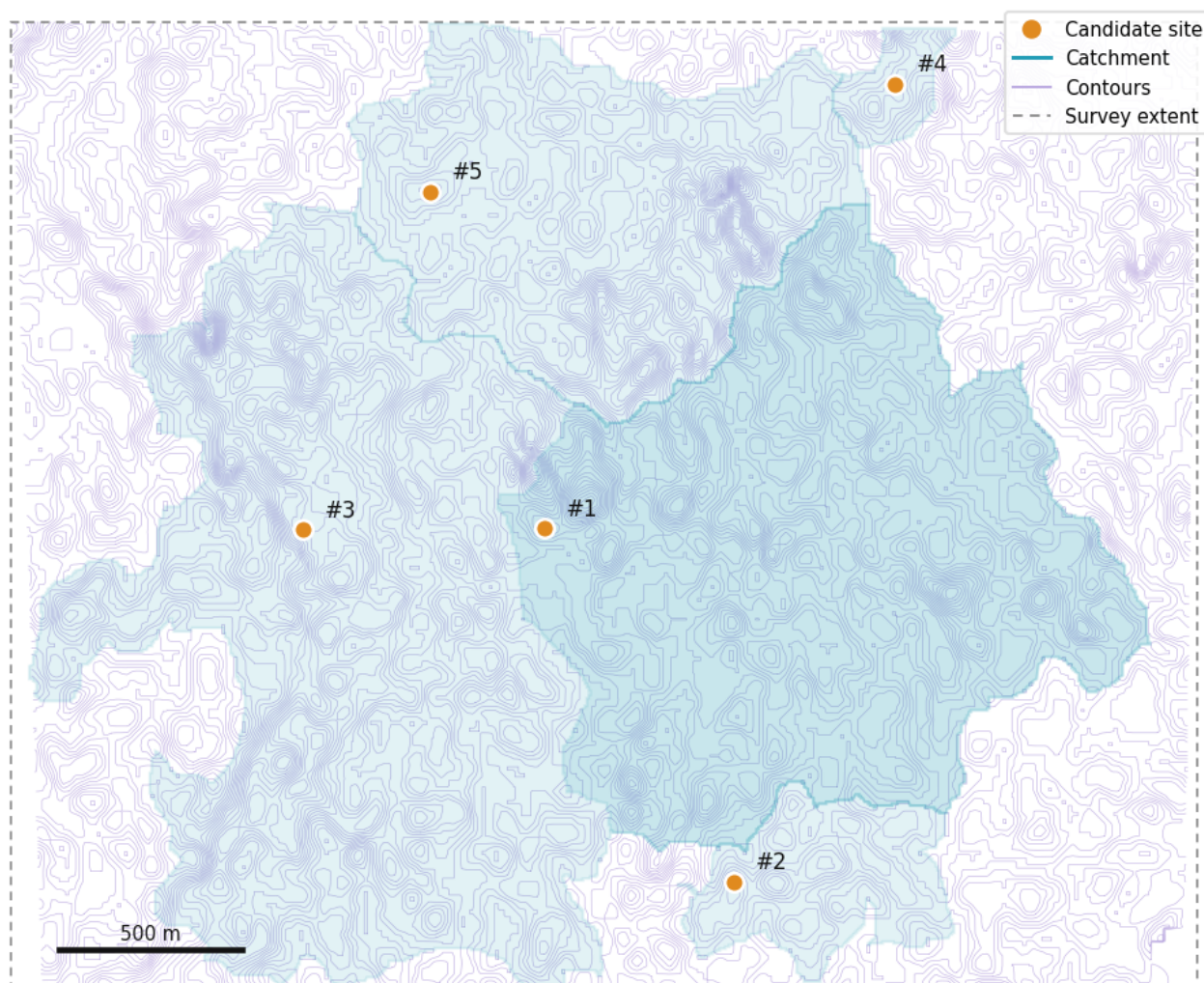
Recommended pond — site #1 of 5 assessed, scoring 81.3/100

4.5 m deep · 141 × 141 m · 81,682 m³

Indicative cost ₹1,20,48,099 at ₹130/m³ excavation. Location 21.25114 N, 81.29545 E.

What limits this pond: practical excavation depth. The pond is at the deepest that is practical to excavate and maintain by the usual methods. Going deeper needs machinery and side protection that change the cost basis, so widening is the cheaper way to add capacity.

Where it goes

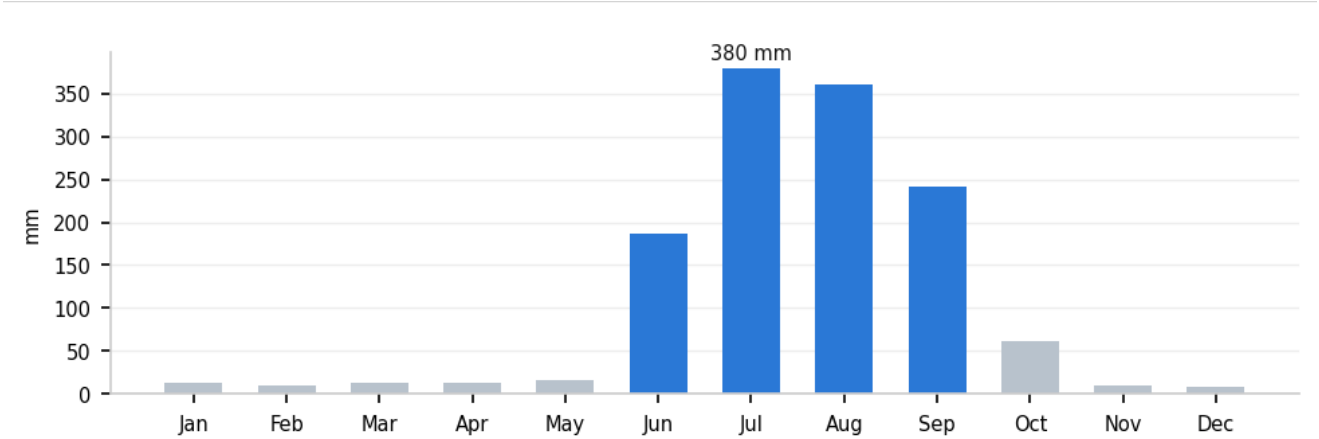


Candidate sites, the catchment draining to each, and the surveyed extent. Contours are the uploaded sheet's own lines. No basemap: the figure is drawn from the analysis so the report can be produced with no network.

The catchment that feeds it

Contributing area	180.25 ha (1.8025 km ²)
Relief	16.01 m
Longest flow path	2836.3 m
Time of concentration (Kirpich)	65 min
Mean slope	3.94 %

Water available



Mean monthly rainfall, 1996 to 2025. Monsoon months (Jun, Jul, Aug, Sep) carry 88.9 % of the annual total. Source: Open-Meteo.

Mean annual rainfall	1312.8 mm
75 % dependable rainfall	1167.9 mm
Coefficient of variation	0.174
Composite curve number (AMC II)	87.7
Hydrologic soil group	D
Annual runoff	599,007 m ³ (C = 0.253)
Design yield at 75 % dependability	418,652 m³

The pond holds 19.5 % of the design-year yield. A pond is not meant to capture all of it — the rest passes downstream, and sizing for the whole yield would mean a structure the parcel cannot hold and a spillway carrying most of the flow anyway.

The other sites assessed

#	Score	Kind	Catchment	Capacity	Limited by
1	81.3	channel position	180.25 ha	81,682 m ³	practical excavation depth
2	72.3	channel position	24.505 ha	22,270 m ³	sustainable yield share
3	72.1	natural depression	381.087 ha	13,929 m ³	practical excavation depth

#	Score	Kind	Catchment	Capacity	Limited by
4	70.9	natural depression	6.55 ha	8,055 m ³	sustainable yield share
5	68.8	natural depression	97.47 ha	81,682 m ³	practical excavation depth

How the recommendation was reached

Sites are scored against 5 criteria weighted by the Analytic Hierarchy Process, then clustered so that two recommendations cannot describe the same structure. The weights below are the ones this analysis used; they are renormalised over the criteria the available data supports, so a missing layer redistributes its weight rather than silently scoring zero.

Criterion	Weight	This site	Contribution
flow accumulation	0.288	0.973	0.28
slope	0.247	0.547	0.135
depression depth	0.192	1.0	0.192
plan concavity	0.11	0.61	0.067
land availability	0.164	0.85	0.14
Score			81.3/100

The contributions sum to the score, so the ranking can be checked rather than taken on trust. The weight vector's own internal consistency is auditable at [GET /api/v1/suitability/weights](#) (Consistency Ratio 0.009, against Saaty's 0.10 threshold).

Assumptions and limitations

- **The terrain is interpolated from contour lines**, not surveyed directly. Between two contours the surface is inferred, so a hollow smaller than the contour interval (1.0 m here) cannot appear in it at all.
- **The pond footprint has no orientation**. The design fixes plan dimensions and depth; nothing in the model chooses a bearing, so the rectangle on the map shows size and position only.
- **Land tenure is not modelled**. No open dataset gives village-level ownership, so a site that is physically ideal may be privately held. This has to be checked on the ground.
- **Existing structures come from OpenStreetMap**, whose coverage varies by village. A building absent from OSM is not evidence of open ground, so exclusions are a lower bound.
- **Costs are indicative**. A single excavation rate is applied to the whole volume; it carries no lead-in, no lining, no land acquisition, and no local rate schedule.
- **Runoff is a design estimate, not a gauged measurement**. SCS-CN with an initial abstraction of 0.35, which is CWC and IMD practice for India rather than the 0.25 of the original American method.

Data sources

Layer	Source	Licence
Terrain	Uploaded contour map (contours_1m.kml)	supplied by the user
Soil	ISRIC SoilGrids — SoilGrids v2.0 (clay/sand/silt, 0-5 cm)	CC-BY 4.0
Land cover	ESA WorldCover — ESA WorldCover 2021 v200	CC-BY 4.0
Rainfall	Open-Meteo — ERA5-Land reanalysis (archive)	CC-BY 4.0, non-commercial use

Analysis 0b306913f4964903 · generated by Contour, an open-data village pond planning tool. Every figure in this report is derived from the uploaded contour map and the sources above; none is a placeholder.